

A Joint Power Approach to Define Countermovement Jump Phases Using Force Platforms

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ABSTRACT

HARRY, J. R., L. A. BARKER, and M. R. PAQUETTE. A Joint Power Approach to Define Countermovement Jump Phases Using Force Platforms. *Med. Sci. Sports Exerc.*, Vol. 52, No. 4, pp. 993–1000, 2020. Conflicting methodologies are used to define certain countermovement vertical jump (CMVJ) phases, which limits the identification of performance-enhancing factors (e.g., rate of force development). **Purpose:** We (a) utilized a joint power approach to define CMVJ phases that accurately describe body weight unloading (i.e., unweighting) and eccentric (i.e., braking) actions, which were combined with the robustly defined concentric (i.e., propulsion) phase, and (b) determined whether the phases can be identified using only ground reaction force (GRF) data. **Methods:** Twenty-one men performed eight maximal CMVJs while kinematic and GRF data were obtained. Hip, knee, and ankle joint powers were calculated by multiplying net joint moments (obtained using inverse dynamics) by joint angular velocities. The net sum of the joint powers (JP_{SUM}) was calculated to define phases by the preeminence of negative (i.e., net eccentric actions) or positive (i.e., net concentric actions) power where appropriate. Unloading, eccentric, and concentric phases were identified using JP_{SUM} and linked to GRF and center of mass velocity features. **Results:** Bland and Altman plots of the bias and 95% confidence intervals for the limits of agreement (LOA), intraclass correlation coefficients (ICC), and coefficients of variation (CV) indicated precise agreement for detecting the unloading (bias, 0.060 s; LOA, -0.110 to 0.229 s) and eccentric (bias, 0.012 s; LOA, -0.010 to 0.040 s) phases with moderate (ICC, 0.578; CV, 40.72%) and excellent (ICC, 0.993; CV, 2.18%) reliability, respectively. The eccentric phase should be divided into yielding (eccentric actions while accelerating downward) and braking (eccentric actions while decelerating downward) subphases for detailed assessments. **Conclusion:** CMVJ phases defined by combining joint and center of mass mechanics can be detected using only force platform data, enabling functionally relevant CMVJ assessments using instrumentation commonly available to practitioners. **Key Words:** JUMPING, CONCENTRIC, ECCENTRIC, GROUND REACTION FORCES, STRETCH-SHORTENING CYCLE

Countermovement vertical jump (CMVJ) performance is often measured during assessments of physical ability because of its positive association with strength, power, and speed (1–3). The CMVJ involves a descent phase followed by an ascent phase leading to airborne flight, and analysis of the phases can provide better insight into a performer's neuromuscular function as compared with the output (i.e., vertical displacement) of the jump (4). However, conflicting methodologies are used to identify important CMVJ phases, which leads to use of identical terminologies (e.g., eccentric)

describing very different time periods within the CMVJ (5,6). This challenges the ability to reliably extract, analyze, and interpret force–time variables often used in contemporary research studies. For the rate of force development, a force–time variable that is critical to jumping performance (7), the definitions for the start and/or end of a specific CMVJ phase dictate whether it is (5,8,9) or is not (10,11) associated with performance differences. Given the value of CMVJ assessments to monitor training program effectiveness (12), compare athlete-to-athlete ability (13–15), and determine physical readiness (16), appropriate phase identification is needed to maximize detection of meaningful force production and performance changes during CMVJ tests.

As early as 1978, Komi and Bosco (17) used the terms “unweighting,” “eccentric,” and “concentric” to identify CMVJ phases using only force platform data. Specifically, unweighting, eccentric, and concentric represented ground reaction force (GRF) production below body weight, deceleration during downward center of mass (COM) motion via negative work, and upward COM motion via positive work, respectively. A recent review suggests that many studies prefer to deconstruct the CMVJ into only eccentric and concentric phases defining the

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periods of downward (i.e., countermovement) and upward (i.e., jumping) COM motion (7), although an initial “unloading” phase has also been used to demonstrate the reduction of vertical GRF to start the CMVJ (5,18). Others have defined phases solely on certain COM kinematic features, with phases defined as (a) negative displacement and negative acceleration, (b) negative displacement and positive acceleration, (c) positive displacement and positive acceleration, and (d) positive displacement and negative acceleration (19,20). However, this approach seems less common presumably because the positive displacement and negative acceleration phase does not involve significant joint rotations or force production contributing to the movement. Importantly, the eccentric and concentric actions of the individual muscles crossing the hip, knee, and ankle joints might not reflect definitions for “eccentric” and “concentric” CMVJ phases based on COM mechanics. This is mostly because muscular direction of pull can oppose the direction of COM acceleration in addition to that of joint angular rotation (21). Recently, use of “unweighting” was recommended alongside “braking” and “propulsion” (4), with the latter two being renamed versions of Komí and Bosco’s (17) “eccentric” and “concentric” phases to circumvent challenges linking individual joint kinetics to COM kinematics. However, these “unweighting” and “braking” definitions do not accurately reflect

what takes place during the CMVJ. This is because the negative net sum of lower body joint power (hip power + knee power + ankle power; JP_{SUM}) indicates that eccentric actions dominate the CMVJ shortly after movement initiation, and this negative JP_{SUM} occurs alongside a positive rate of vertical GRF development (Fig. 1A). Because these actions occur during “unweighting,” it is clear that eccentric braking actions occur before the aforementioned definitions of COM “braking” (22).

The combination of GRF and motion capture data enables the calculation of angular power production about the hip, knee, and ankle joints. These joint powers can be summed to create the net sum of joint power (JP_{SUM}) and indicate whether the joints are predominantly producing positive (i.e., concentric) power or negative (i.e., eccentric) power (23,24). By combining joint powers with COM mechanics for this purpose, CMVJ phase definitions can be refined to describe both the dominant muscle actions (e.g., eccentric or concentric) throughout the lower body and the external COM mechanics (e.g., braking) where appropriate. Considering the impracticality of advanced motion capture instruments and data analysis techniques in practitioner settings, it is important that the CMVJ phases defined based on JP_{SUM} can be identified using GRF data alone because of the practicality and widespread use of force platforms for

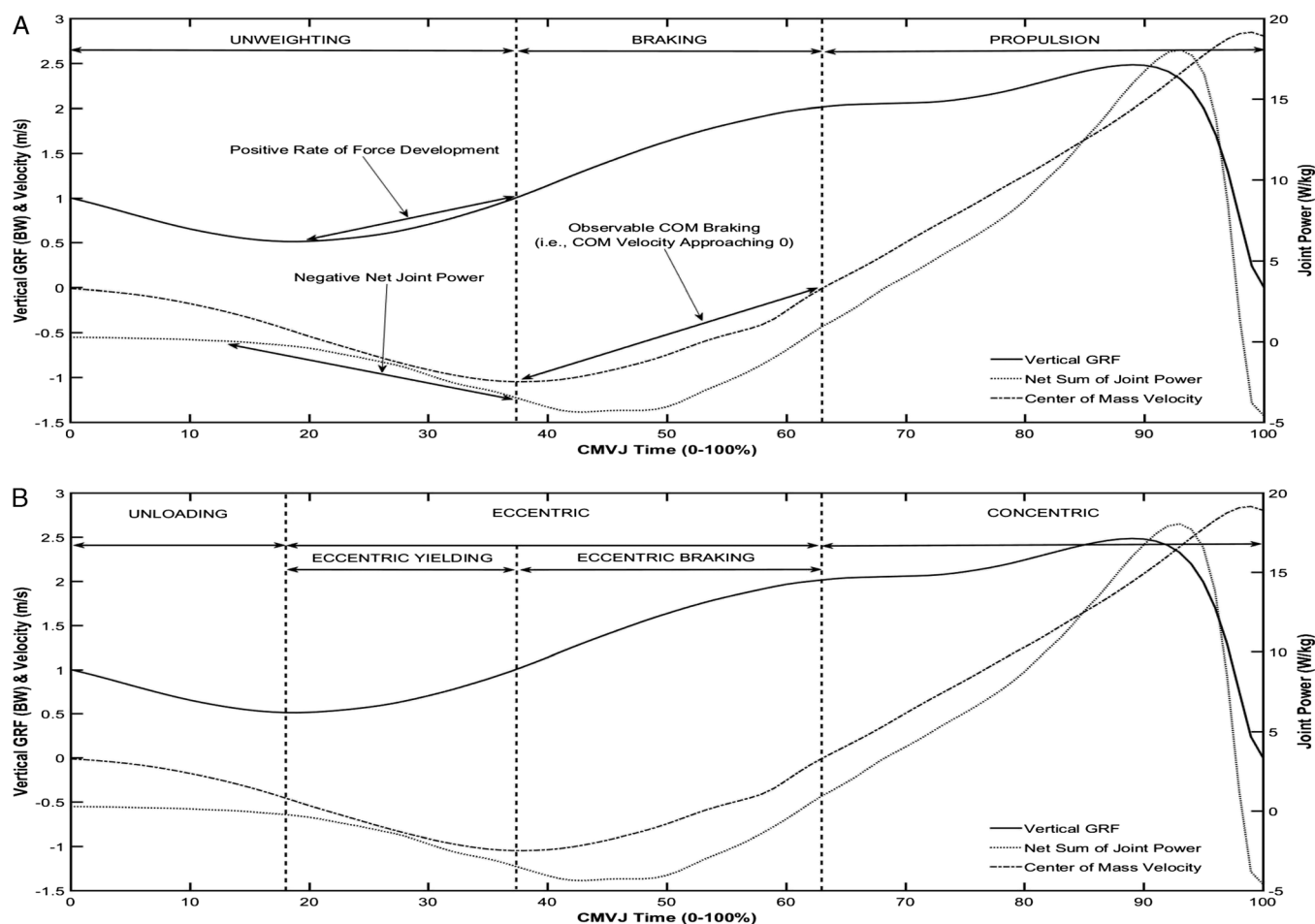


FIGURE 1—Group mean ensemble curves for vertical GRF, vertical COM velocity, and the net sum of joint power during the CMVJ with commonly recommended phases alongside features showing the challenges of the definitions (A) and the proposed phases that address those challenges (B).

performance tests (25). This would achieve the following aims: (a) establish CMVJ phases that describe both the net lower limb muscle action and the external COM mechanics, and (b) facilitate use of the phases in practitioner settings without the need to acquire motion capture data acquisition devices. The purposes of this study were to (a) utilize a JP_{SUM} approach to define CMVJ phases that isolate body weight unloading (i.e., unweighting) and eccentric actions (i.e., braking), and (b) determine whether the proposed phases can be accurately identified using only data from force platforms (e.g., vertical GRF and COM velocity). We hypothesized that specific CMVJ phases related to net muscular action could be determined using the JP_{SUM} approach, and that CMVJ events from the JP_{SUM} approach could be accurately identified using solely vertical GRF and COM velocity data.

METHODS

Subjects. Twenty-one recreationally active (self-reported twice-weekly participation in recreational sports or activities requiring the CMVJ for at least 6 months before participation) men (26 ± 3 yr, 85.68 ± 14.79 kg, 1.79 ± 0.08 m) were examined in this study. All were free of any prior injuries or ailments that would impair their CMVJ abilities. Written informed consent was obtained in accordance with institutional review board requirements before all laboratory activities.

Experimental protocol. Participants wore laboratory footwear (Vazee Pace v2 or Fresh Foam Veniz; New Balance Athletics, Inc., Boston, MA) and completed a ~10-min warm-up, consisting of their choice of 5 min of stationary cycling or walking/jogging on a treadmill at a self-selected intensity followed by CMVJs progressing in intensity from moderate to maximum effort. Participants requiring a more vigorous warm-up also performed body weight squats, additional submaximal CMVJs, and stationary lunges. Kinematic and kinetic data were obtained using a 12-camera three-dimensional (3D) motion capture system (Vantage v5 and T40 series cameras; Vicon Motion Systems, Ltd., Oxford, United Kingdom; 200 Hz) and two 3D force platforms (Kistler Instruments, Corp., Amherst, NY; Advanced Mechanical Technology, Inc., Watertown, MA; 1000 Hz). Segmental motion was tracked using spherical 14-mm reflective markers adhered over the following locations using hypoallergenic adhesive tape: anterior–superior iliac spines, posterior–superior iliac spines, iliac crests, sacrum, medial and lateral aspects of the knee joint, medial and lateral malleoli, and on the shoe over the base of the second toe. In addition, plastic shells with four noncollinear reflective markers were adhered to the lateral aspect of the thigh and leg segments using elastic wraps and hypoallergenic adhesive tape, whereas three noncollinear reflective markers were adhered over each shoe's heel counter. After a static calibration trial, markers adhered over the anterior–superior iliac spines, medial and lateral aspects of the knee joint, medial and lateral malleoli, and base of the second toe were removed, and the remaining markers were used for motion tracking during eight CMVJ trials. Participants began each trial standing still with each foot on a force platform. Participants were instructed

to complete the CMVJ using preferred countermovement depths and arm swings to target maximal jump height and the shortest possible jump time. Participants were instructed to land with each foot contacting a force platform before returning to a standing still position. Up to 2-min rest was provided between trials.

Data analysis. Raw data were imported to Visual3D (version 6; C-Motion, Inc., Germantown, MD). A five-segment model was constructed from the raw marker trajectories to include the pelvis and right thigh, leg, and foot segments. Raw kinematic and kinetic data were interpolated using a third-order polynomial with a maximal gap of 10 data points, although the gaps identified in the collected data, when present, were much smaller than 10 data points. Data were then smoothed using a four-order low-pass Butterworth digital filter with cutoff frequencies of 4 and 30 Hz, respectively. Cutoff frequencies were determined by transforming the raw kinematic and kinetic data from the time domain to the frequency domain and visually inspecting where the majority of signal resided. Despite claims that kinematic and kinetic data should be filtered using the same cutoff frequencies (26,27), the rationale for the procedure remains a topic of debate (28) and potential benefits of the procedure have only been discussed relative to impact movements with very rapid loading rates (e.g., sidestep cutting and landing).

GRF data from the two force platforms were summed along the vertical axis to create a vertical GRF acting at the body COM. The position of the body COM was represented by the pelvis COM, and COM vertical velocity was calculated as the derivative of the vertical COM position, although it can also be obtained from vertical GRF data by calculating vertical acceleration using Newton's law of acceleration and then obtaining the time integral of the acceleration. We chose to use the pelvis COM to reduce the number of calculations needed to obtain a representation of the COM while using a valid procedure to track COM motion during the CMVJ (29). A Cardan (X, Y, Z) rotation sequence and the right-hand rule were used for 3D joint angle computations, where X represents the medial–lateral axis, Y represents the anterior–posterior axis, and Z represents the longitudinal axis. The angles of the hip, knee, and ankle joints were defined as the thigh relative to the pelvis, leg relative to the thigh, and foot relative to the leg, respectively. Newtonian inverse dynamics procedures and the right-hand rule were used for 3D net internal joint moment calculations, with moments being resolved in the coordinate system of the proximal segment. Joint angular powers were calculated as the dot product of the net joint moments and joint angular velocities, with the latter calculated as the derivative of the joint angular positions. Data were trimmed to the time between the start of the CMVJ (i.e., vertical GRF < 97.5% body weight) (30), where body weight was calculated as the average vertical GRF during the first 500 frames of quiet standing, and takeoff (i.e., vertical GRF < 20 N), time normalized to 101 data points (0%–100%), and exported to MATLAB (R2017a; The Mathworks, Inc., Natick, MA). Although we determined the criteria for the start of the CMVJ according to the results previous work (30), it is possible to identify this event using a threshold equal to the mean of vertical GRF during a period of quiet standing ± 5 SD (31) or a threshold above or below 1.75 times body

weight and using a backward search to locate the time when vertical GRF passed through body weight (32).

Mean ensemble curves for the time-normalized vertical GRF, vertical COM velocity, and hip, knee, and ankle joint power were calculated, and JP_{SUM} (sum of time-normalized hip, knee, and ankle joint power curves) was obtained to determine the dominant muscle action (positive JP_{SUM} : concentric; negative JP_{SUM} : eccentric) occurring across the lower body. Onsets of negative and positive JP_{SUM} defined the end points for net eccentric and concentric CMVJ phases. Vertical GRF and/or COM velocity phase identifiers (5,6,33) were temporally linked to the onsets of negative and positive JP_{SUM} (i.e., net eccentric and concentric actions, respectively). Three main phases, referred to as “unloading,” “eccentric,” and “concentric,” were identified based on temporal alignments of the onsets of negative and positive JP_{SUM} and the vertical GRF and/or COM velocity identifiers (Fig. 1). Importantly, the eccentric phase was divided into “yielding” and “braking” subphases based on the fact that performers produce a negative JP_{SUM} , whereas the COM velocity accelerates then decelerates in a downward direction, indicating that unique movement effects (i.e., COM yielding and braking, respectively) occur while producing a negative JP_{SUM} (i.e., net eccentric actions). Table 1 describes the phase definition for each method in addition to practical descriptions of what the phases represent functionally. Terminology was selected to maintain consistency with relevant vertical GRF approaches to identify one or more of these phases (5,17,18).

Agreement between the joint power and vertical GRF approaches was examined using Bland and Altman plots and a limits of agreement (LOA) approach, in which agreement was determined by the bias (i.e., offset) of the difference between methods (34). LOA values were calculated, and 95% confidence intervals (CI) for the LOA were used to account for uncertainty in the estimates (35) and provide the precision

of the estimated agreement between the two approaches (34). The frequency distribution for the difference between methods was used to visually determine normality. We could not determine an *a priori* literature-based criterion for adequate agreement (i.e., bias) because of a lack of similar investigations. Thus, we defined agreement using a bias threshold of ± 0.08 s, which is $\sim 20\%$ to 25% of the typical unloading and eccentric phase durations observed using the current GRF-based definitions (18,22). Root mean square error (RMSE) was calculated for agreement comparisons with future studies. SPSS (version 25; IBM Corp., Armonk, NY) was used to determine intraclass correlation coefficients (ICC; model: two-way random; type: absolute agreement) and the corresponding 95% CI to supplement to the 95% CI for the LOA with respect to relative reliability. ICC values were interpreted as follows: poor $< 0.5 \leq$ moderate $< 0.75 <$ good $\leq 0.9 <$ excellent (36). In addition, the coefficient of variation (CV), as described by Glüer and colleagues (37), was calculated to provide additional reliability information.

RESULTS

Figures 2 and 3 present the agreement between the JP_{SUM} -based and vertical GRF/COM velocity-based methods to identify the unloading and eccentric phases, respectively. Table 2 provides the bias and 95% CI for the LOA (i.e., random error) between approaches in addition to RMSE, ICC, and CV statistics. The onset of negative JP_{SUM} , which defined the end of the unloading phase, was precisely identified using the local minimal vertical GRF, as evidenced by a bias (0.06 s) below the ± 0.08 -s threshold. Reliability between approaches defining the end of the unloading phase was moderate according to the ICC and CV data. In addition, the onset of positive JP_{SUM} , which defined the end of the eccentric phase, was precisely identified using the $0\text{-m}\cdot\text{s}^{-1}$ vertical COM velocity threshold, as evidenced

TABLE 1. Current methods used to identify the proposed phases and subphases in addition to practical descriptions of the phases to aid in research applications and training prescriptions.

Phase	Start of Phase		End of Phase		Practical Phase Description
	GRF Method	JP_{SUM} Method	GRF Method	JP_{SUM} Method	
Unloading	Vertical GRF $< 97.5\%$ BW (22)	Vertical GRF $< 97.5\%$ BW	Local minimal GRF (5)	Onset of negative JP_{SUM}	Body weight is unloaded to begin the countermovement by converting gravitational potential energy to downward kinetic energy (i.e., starting free fall). This may be facilitated by primarily positive hip joint power, but this remains to be confirmed.
Eccentric yielding	Local minimal GRF	Onset of negative JP_{SUM}	Peak negative COM velocity	Peak negative COM velocity	Positive rate of force development occurs through predominantly negative lower limb joint power (i.e., eccentric muscle actions) to oppose gravity and stop downward motion. Despite these efforts, gravity is not yet resisted as the COM continues downward acceleration.
Eccentric braking	Peak negative COM velocity	Peak negative COM velocity	COM velocity $> 0\text{ m}\cdot\text{s}^{-1}$ (4,5,13)	Onset of positive JP_{SUM}	Positive rate of force development through predominantly negative lower limb joint power (i.e., eccentric muscle actions) continues to occur to oppose gravity and stop downward motion. During this time, gravity is being resisted and downward COM velocity slows down.
Concentric	COM velocity $> 0\text{ m}\cdot\text{s}^{-1}$	Onset of positive JP_{SUM}	Vertical GRF $< 20\text{ N}$	Vertical GRF $< 20\text{ N}$	Positive joint power and concentric actions dominate the output of the lower extremity to propel the COM as quickly as possible. The jumper has overcome gravitational resistance.

BW, body weight.

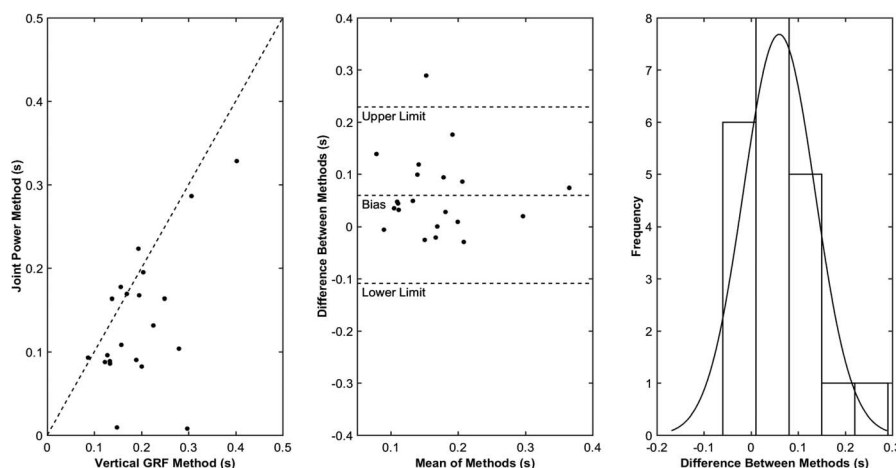


FIGURE 2—Bland and Altman assessment of agreement between joint power and vertical GRF methods to define the end of the unloading phase. Note: left plot illustrates the end of the unloading phase determined by both methods with a dashed line of equality; middle plot illustrates the Bland and Altman plot with the bias (offset from zero line) and upper and lower 95% CI for the upper and lower LOA (random error); right plot illustrates the frequency distribution for the difference between methods.

by a bias (0.012 s) well below the ± 0.08 -s threshold. Reliability between approaches for defining the end of the eccentric phase was excellent according to the ICC and CV data.

DISCUSSION

The purposes of this study were to (a) utilize a JP_{SUM} approach to establish CMVJ phases that accurately define body weight unloading (i.e., unweighting) and eccentric braking actions, and (b) determine whether the proposed phases can be accurately identified using only force platform data. Our purposes were based on the fact that the “unweighting” and “braking” phases recently recommended (4) do not accurately reflect unweighting (i.e., unloading) of body weight or the braking initiation actions that occur via lower body eccentric muscular involvement before observable slowing of the

COM. Our findings indicate that CMVJ phases determined using the JP_{SUM} approach can be accurately and reliably detected using only force platform data (i.e., vertical GRF and COM velocity derived from the pelvis COM). Given the fact that force platforms have been among the most commonly used biomechanical instruments in human performance research and practice (4) for decades (25), it is important to note that this study indicates that the proposed phases can be used to deconstruct the CMVJ for analyses in testing environments where more advanced instrumentation (e.g., motion capture systems) is not readily available.

Importantly, the onset of the negative JP_{SUM} after the start of the CMVJ ($\sim 17\%$ normalized time) was closely aligned with the local minimal vertical GRF ($\sim 19\%$ of normalized time). Therefore, “unloading” or “unweighting” ends at the local minimal vertical GRF when the lower body extensors predominantly produce eccentric actions. Because the time

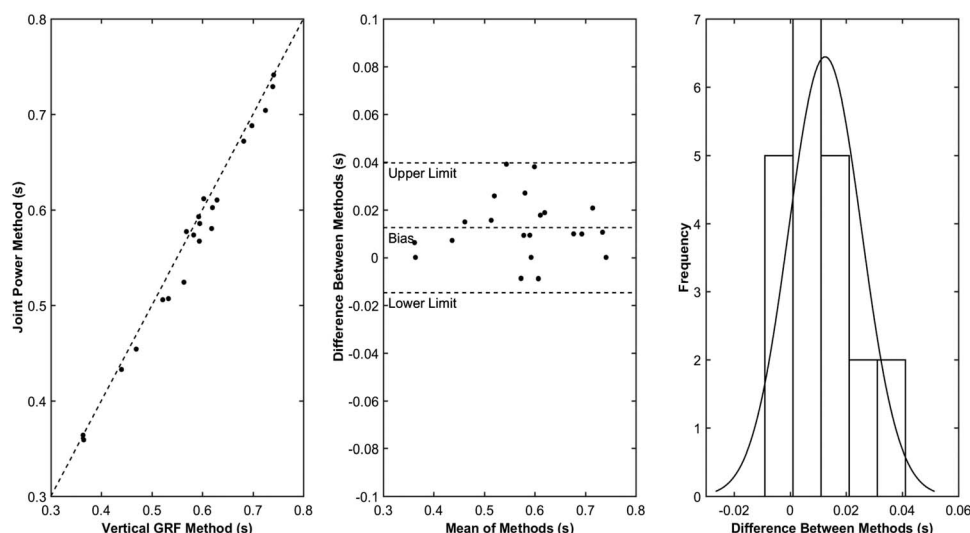


FIGURE 3—Bland and Altman assessment of agreement between joint power and vertical GRF methods to define the end of the eccentric phase. Note: left plot illustrates the end of the eccentric phase determined by both methods with a dashed line of equality; middle plot illustrates the Bland and Altman plot with the bias (offset from zero line) and upper and lower 95% CI for the upper and lower LOA (random error); right plot illustrates the frequency distribution for the difference between methods.

TABLE 2. Bias (offset), random error (95% LOA lower and upper LOA), RMSE, and reliability statistics between methods to identify CMVJ phase end points.

Phase End Point	Bias	Lower 95% LOA	Upper 95% LOA	RMSE	ICC (95% CI)	CV
Unloading	0.060	-0.110	0.229	0.096	0.578 (-0.023 to 0.710)	40.72
Eccentric	0.012	-0.010	0.040	0.018	0.993 (0.942 to 0.998)	2.18

All variables reported as units of time (in seconds) except the CV, which is a percentage.

between the start of the CMVJ and the local minimal vertical GRF has been called the unloading phase (5,18,22), we chose to continue use of that term over the “unweighting” term. From a practical perspective, the local minimal vertical GRF magnitude is correlated with enhanced CMVJ performance (5), and faster unloading times were shown to separate good from poor recreational jumpers when performance was defined by the modified reactive strength index and jump height (18). Strategic cues or training methods leading to greater unloading and/or faster unloading times are currently unknown, but such interventions should be examined to reveal methods for increasing downward kinetic energy to influence eccentric demands. Velocity-based training using external loading could be studied for this purpose because greater unloading should coincide with faster downward velocities and therefore increased eccentric force production demands during the later phases if appropriately controlled to the performer’s preferred countermovement depth.

After unloading, the negative JP_{SUM} (i.e., net eccentric action) coincides with the increase in the vertical GRF until ~64% of normalized time, at which point JP_{SUM} becomes positive (i.e., start of net concentric actions). This indicates that the “eccentric” phase is between the end of unloading and the onset of positive JP_{SUM} . During this “eccentric” phase, the performer is applying force into the ground to try and stop downward COM acceleration and velocity, thereby terminating the countermovement and propelling the COM upward. Importantly, the periods of accelerating and decelerating negative COM velocity within this eccentric phase occur when the vertical GRF displays a positive rate of change. This indicates that participants are temporarily yielding to the force of gravity from the end of unloading until ~38% of normalized time because the negative JP_{SUM} and increasing force production does not decelerate the COM. This demonstrates that the initiation of “braking” via eccentric muscle actions (i.e., joint mechanics) begins before positive COM acceleration (i.e., external mechanics). These braking initiation actions are analogous to the initial pressing of the brakes in a car that is accelerating downhill. The initial pressing of the brakes and the requisite response of bringing the car’s acceleration closer to zero are separated by a brief period of time. Thus, pressing the brakes does not immediately slow the car, but it is a critical first step to the process. To describe both muscular effort (negative JP_{SUM} ; net eccentric actions) and the COM movement qualities (i.e., yielding to gravity), we coined this time period the “eccentric yielding” subphase of the main eccentric phase (Fig. 1B).

Greater rates of yielding GRF seem to contribute to enhanced CMVJ performance via greater force production during subsequent phases (14) or overall explosiveness (38). Prior work (39–41) suggests that CMVJ performance can

improve through a rapid reduction of muscle slack (i.e., uptake of slack in the fascicles and tendon tissues) and achievement of active state (maximized number of actin binding sites available for cross-bridge formation) during the countermovement. We presume that vertical GRF production during eccentric yielding represents the time when the uptake of muscle slack facilitates the transition to active state. Thus, more rapid force production during eccentric yielding to begin the braking process could reflect a refined ability to quickly take up muscle slack and achieve active state, ultimately facilitating a more efficient transfer of force to the bones before observable COM deceleration and change of direction. From a practical perspective, eccentric-focused power exercises, such as loaded jump squats (12), could enhance braking initiation abilities (i.e., rate of force development during yielding) if downward kinetic energy is not compromised by slow countermovement velocities. Plyometric activities that increase downward kinetic energy before yielding occurs (e.g., drop jumps, CMVJs with weighted vests, etc.) may also stimulate yielding adaptations.

The onset of positive JP_{SUM} was very closely aligned with the onset of positive vertical COM velocity (~65% of normalized time; Fig. 1). The time between the end of eccentric yielding and the onset of positive JP_{SUM} is nearly identical to the “eccentric” phase described by Komi and Bosco (17), and the “braking” phase recently promoted by McMahon and colleagues (4). Therefore, we defined this time as the “eccentric braking” subphase of the main eccentric phase (Fig. 1B). Key considerations for eccentric braking center on the fact that the amount of force production (i.e., impulse) during this subphase must equal the impulse during the unloading and eccentric yielding periods combined (4,19). Achieving this impulse magnitude rapidly is heavily dependent on the strategy of the jumper. However, the vertical GRF magnitude achieved at the end of the phase, which represents the amount of stored elastic energy in the system (42), must be sufficiently high to quickly complete the CMVJ (5) and maximize mechanical output during the concentric phase of the CMVJ (22) through increased positive work performed.

Deconstructing the CMVJ to isolate the eccentric subphases can help researchers and practitioners determine whether jumpers are able to capitalize on the benefits of rapid GRF production, or address slow GRF production, while yielding to gravity. Similar to the eccentric yielding subphase, unloaded or loaded plyometric exercises (depth jumps, loaded CMVJs, or jump squats, etc.) with rapid changes of vertical direction (12,43) could be used to stimulate eccentric braking adaptations as long as downward velocity induces sufficient kinetic energy. In addition, general strength training with high-intensity loads that require quick deceleration (e.g., squats) could also develop eccentric braking force production abilities (44). The goal for braking-specific training should be to produce the greatest possible GRF magnitude at the end of the eccentric phase within the shortest duration.

The period of time after the end of the eccentric phase has been referred to as both “concentric” (5,6,17) and “propulsion” (4). However, to accurately link the dominant muscular action to the CMVJ phases where possible and ensure use of a consistent

naming convention with much of the reviewed literature (7), we defined this time period as the “concentric” phase. Importantly, when targeted adaptations are achieved during the eccentric sub-phases, concentric variables that can predict CMVJ performance, such as peak knee and ankle joint power (45,46) and peak power production (12), should increase alongside the statistical and mechanical determinant of CMVJ performance, impulse (47), to produce a higher jump. As such, it may be more advantageous to directly target eccentric adaptations through externally loaded training (see previous examples) compared with direct targeting of concentric adaptations. Nonetheless, it is generally recommended that all concentric actions be performed with maximal intent during training.

A possible limitation of this study was the moderate reliability and relatively large bias and RMSE values associated with the agreement between methods for detecting the end of the unloading phase. This result was due to two participants exhibiting a negative JP_{SUM} almost immediately after initiating the CMVJ (Fig. 2) because of an increase in vertical GRF production before the start of unloading. When these two participants were removed and agreement statistics were rerun, agreement and reliability greatly improved (bias, 0.044 s; lower 95% LOA, -0.070 s; upper 95% LOA, 0.162 s; RMSE, 0.068 s; ICC (95% CI), 0.769 (0.207–0.921)); CV, 28.21%). This indicates that a potential scenario in which the described methods for defining the end of unloading are not likely to agree is when participants apply force into the ground and move the arms and COM upward before the vertical GRF decreases below the 97.5% body weight threshold. If these initial actions before unloading are considered to be a natural strategy and therefore critical to the CMVJ analysis, the start of the jump can be defined using the alternative procedures described previously in the methods. However, we argue against these procedures, as they permit a load–unload strategy that is not commonly observed in our laboratory, and defining the start of an unloading phase using a loading action can weaken the link between the name of the phase and its functional and mechanical characteristics. Given these observations, restricting arm swing might be a beneficial control strategy if the JP_{SUM} approach is utilized because it should be less likely for unusual unloading strategies to be employed as a result of participant-specific or trial-to-trial arm swing techniques. Thus, studies adopting the unloading phase are encouraged to use the local minimal vertical GRF instead of the onset of negative JP_{SUM} to determine the end of the phase and avoid potentially premature detection of the end of the phase.

Our focus was to establish functionally and mechanically relevant CMVJ phases that could be defined solely using force platform data, and we suggest defining the phases using the

described GRF method because of the practical use of force platforms in human performance environments (4,25). However, given the potential use of low-cost, portable kinematic-only tracking devices to monitor physical performance, such as depth-tracking cameras and human pose estimation (48), it is important to consider how the phases can be detected using such instruments. For instance, given use of the current CMVJ performance instructions and the presence of a pose landmark (i.e., pelvis, hip, etc.) representing the body COM, velocity and acceleration can be obtained as the first and second derivatives of the pose COM landmark’s position, respectively. The start of the CMVJ could then be defined as a 2.5% reduction in a force curve estimated using Newton’s law of acceleration (i.e., multiplying the COM acceleration by participant mass). The end of unloading would be defined by the local minimum COM vertical acceleration. The eccentric phase could then be defined as described in Table 1 under the vertical GRF method by using the velocity of the pose COM landmark. Finally, the end of the concentric phase could be defined by the data point immediately before the time when the vertical acceleration of the COM landmark becomes equal to gravity ($-9.81 \text{ m}\cdot\text{s}^{-2}$). It should be noted that using the proposed CMVJ phases with only kinematic data will not identify phases as precisely as when using force platform data or both kinematic and force data. The process outlined in this paragraph simply represents a process for obtaining the best possible phases with kinematic data.

In summary, the CMVJ contains three main phases best defined as “unloading,” “eccentric,” and “concentric.” Precise agreement and reliable measurements between JP_{SUM} and vertical GRF and COM velocity phase end points indicate that force platform data alone can accurately identify the phases. The eccentric phase should be separated into “eccentric yielding” and “eccentric braking” subphases for more thorough assessments of eccentric performance. We recommend use of the proposed phases to increase methodological consistency among future CMVJ studies relative to variable extraction. This will enable researchers and practitioners to improve the link between the muscle actions, whole body motion, and vertical GRF production associated with CMVJ performance.

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The authors have no conflicts of interest to disclose. The results of the present study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

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